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Making Letters Distinctive

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Small graphic changes made in normal letters of the alphabet changed the similarity relations among those letters. Kindergarten, first-grade, and second-grade children as well as adults classified letters of this distinctive font faster and with fewer errors than they classified normal letters. It was shown that it is not features alone but relations between features within letters, and relations between letters in the stimulus set, that determine how difficult any particular letter is to classify. The advantage of the distinctive font is such that many children had less difficulty classifying distinctive letters into bins labeled with normal letters than doing the conceptually easier match-to-sample task of placing normal letters into bins labeled with normal letters.

Many different type styles have been used in books designed to teach children to read. Over the past century or so, these type styles have tended to change from relatively ornamental and heavily serifed letters to "clean" and less cluttered fonts (Lewis, 1963; Tschihold, 1967). A serif is a line or bar at the end of the main strokes in a typed letter. Figure 1 displays examples of such fonts.

New fonts have often been introduced with no real evidence concerning their effect on reading. Rather, the movement toward a clean font has generally been motivated by the aesthetic concerns of some printers and artists and by the intuitions of some educators and psychologists. Printers and artists have considered that letter forms should be in agreement with other artistic trends of the time, whereas educators and psychologists have considered that simple letters, those without serifs or other potentially distracting features, would be easier for children to learn initially and to read. DeVinne (1900) noted that

> A letter of modern cut is really not so distinct as the same letter in the old style. The old punch cutter and the modern punch cutter worked to reach different ends. The old cutter put readability first; he would make his types graceful if he could, but must first of all make them distinct and readable in a mass. His object was to aid the reader. The modern punch cutter thinks it his first duty to make every letter of graceful shape. (p. 190)

This trend, seen 80 years ago by DeVinne, persists today, even though the few empirical studies that have explored the legibility of different type styles have most often concluded that serifed and other slightly cluttered styles are actually easier to read than are the more modern clean styles.

Crossland and Johnson (1928), in studying the identification of single letters by adult observers, concluded that "seraphed letters are more legible than unseraphed letters" (p. 121). Robinson, Abbamonte, and Evans (1971) based a speculation as to why sans serif print styles are less readable than serifed print styles on Dowding's (1957) report that adults take 7½% more time to read the same passage when printed in sans serif type than when printed in serifed type. Paterson and Tinker (1940) studied the legibility of 10 different font styles and reported that 1 very ornamental style was difficult to read. But they also reported that the only sans serif style measured ranked next to last of the 10 fonts in judged legibility and that the style slightly retarded reading speed.

Although these studies were all done with adults, the conclusions appear relevant to

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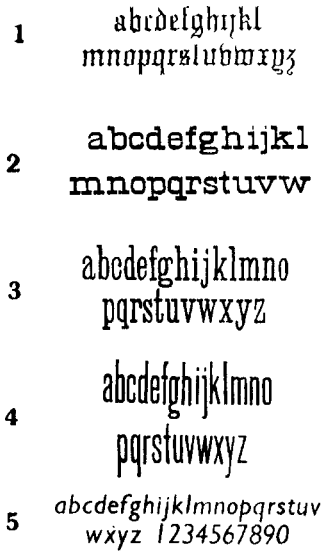

Figure 1. Sample fonts demonstrating the decrease over time in the intensity of serifs added to the basic letters. (1 = Saxon Black, popular in Europe and eighteenth century America; 2 = Egyptian Number 2, popular in the mid-1800s; 3 = Skeleton antique, also popular in the 1800s; 4 = Gothic special, used often in the early 1900s; 5 = Gill sans medium, used currently in many elementary school programs.)

children as well. Burt, Cooper, and Martin (1955), in a study involving 300 children aged 10 and 11, measured reading speed and comprehension when passages were printed in several different fonts. Their investigation did not include a sans serif font because their preliminary work had found that this style was much inferior to the older styles of type face for children. Indeed, they found that even those modern type styles with less prominent serifs were less readable than the older styles:

> The modern type faces were the least legible. As the reader will see if he compares 'n' and 'u' in (say) Garamond and Bodini, respectively, the older faces accentuate those parts that are different, while modern faces accentuate those parts that are similar. With the latter, therefore, letters are far more readily confused. (Burt et al., 1955, p. 33)

These reports all show that adults and children who know the alphabet read less effectively with the modern fonts than with older type faces. Apparently, flourishes on letters can facilitate reading; we have not uncovered any studies showing otherwise. These results all concern the difficulty of reading. However, clean fonts are used most often in materials designed to teach children to read. Therefore, our consideration in this article is that serifs and other flourishes may also be important to children when they are learning the alphabet. Perhaps such characteristics as those that Burt et al. (1955) stated are important for reading are also important when young children classify and identify letters. Perhaps the current trend toward clean print in the primary years enhances confusions and makes it more difficult for children to distinguish, and thus to learn, letters of the alphabet.

This hypothesis, which suggests that slight modifications of letters can result in substantial changes in the ability of children to work with these letters in a classification task, is tested on a very limited portion of the alphabet. For the purpose of the experiments presented here, the letters *b, d, p,* and *q* were primarily used. These are the letters noted by Mach (1897) as being particularly difficult for children to learn, an observation that has received much empirical support over the years (Davidson, 1934, 1936; Popp, 1964; Teegarden, 1932; Weschler & Hagin, 1964). The concepts employed here, however, can be applied equally well to the entire alphabet.

Experiment 1: Adding a Common Feature to Letters

Burt et al. (1955) considered that letters are confusing when the parts of them that are similar to other letters are accentuated. This is consistent with the current idea that letters having the same features are frequently confused and that the more features letters share, the more often those letters are confused with one another (Gibson, 1965; Thorson, 1976; Gibson, Osser, Schiff, & Smith, Note 1). Our interpretation of these opinions is not the interpretation that is usually reported. Perhaps common features, that is, particular common strokes such as curved or straight lines, do not produce the confusions. Rather, relations among such features may be what are important. For example, the fact that *u* and *n* are often confused might not be because they

have common features per se but because the perceived relations between those features are identical. Stated differently, the theoretical problem may be in understanding just what a feature is for the reader. A feature is often thought to be a single stroke of the pen such as a curved or straight line (Geyer & DeWald, 1973; Lindsay & Norman, 1972; Gibson et al., Note 1). However, we suggest that a feature might be the relations among strokes. If a maximally legible alphabet is to be designed eventually, then it is important to decide this question.

Because this point concerning number of common features versus feature relations is essential to our thinking in developing a legible and easy-to-learn font, we examined this issue first. We added the identical feature, a simple dot, to two letters. The letters selected for study were the lower case *p* and *b*. A san serif style was selected for simplicity. The added dot was placed an equal distance from the bottom of each letter. These normal and altered letters are shown in Figure 2. The letters *p* and *b* are often confused by young children. Those who suggest that the confusion is the result of the common features shared by the letters would argue that the dotted letters should be as difficult or more difficult for children because there are more common features in those letters than in the normal letters. Alternatively, since the added dot produces different relations between the letters, those letters should be less similar to each other than are the normal letters. Indeed, the entire sample of 14 adults who were tested rated the dotted letters of Set B less similar to one another than the normal letters of Set A. By this argument, the dotted letters should be more readily discriminated or distinguished from one another (see Monahan & Lockhead, 1977).

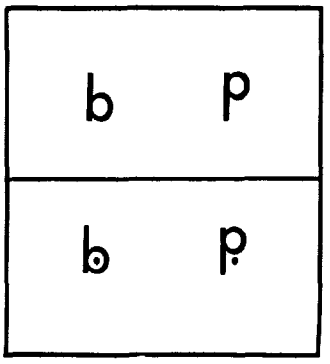

Figure 2. The normal (Set A, top) and dotted (Set B, bottom) letters used in Experiment 1.

Thus, two theories make different predictions concerning the letters in Set A versus those in Set B. Relational predictions state that items which are more perceptually dissimilar are more readily classified; accordingly, the letters with dots should be easier to discriminate from one another than those without dots. Feature-analytic approaches make the opposite prediction. According to overlap-of-features measures (Gibson et al., Note 1), the letters with dots should be at least as difficult to discriminate from one another as those without dots. This is because the proportion of features in common is greater when the same feature, the dot, is included in every stimulus. The intuition that less cluttered letters are easier for children must also predict that the letters of Set A are preferred. Experiment 1 was conducted to test these predictions.

Method

Subjects. All members of a kindergarten class (n = 23) from a public elementary school were tested individually in late October on a match-to-sample task.

Procedure. A small wooden rectangular board with four cut-out bins was placed in front of the child. Each bin was labeled with a "sample" card placed directly above the bin. The child was given a deck of cards and was instructed to place each card from the deck with the sample cards above the bins. Each card was 2 × 3 inches (5 × 7.6 cm) in size and had an individual letter printed in the center.

To assure that each child understood what was expected, a warm-up task was given first. Using the letters *z*, *k*, *l*, and *m*, four sample cards were randomly placed above the bins. A randomly arranged deck of eight cards, two each of the four letters, was used. The experimenter placed the first two cards in the appropriate bins and then handed the remaining deck to the child for completion of the task. Every child quickly understood the nature of the match-to-sample task and performed the warm-up task accurately.

Each child then performed the match-to-sample task, using two letters, *b* and *p*, with the instructions to perform the task as quickly and accurately as possible. A timer was started by the experimenter when the first card was placed into any bin, and the timer was stopped when the last card entered a bin. The elapsed time was recorded as the sorting time. Errors were then determined by tallying the number of cards of each letter that

had been placed in each bin. The center two bins of the board were used; the bins were labeled *b* and *p* for some children and were labeled in the opposite orders for others. There were 20 cards in each set, 10 cards of each of two letters, in the thoroughly shuffled deck. Twelve children sorted the letters of Set B first (see Figure 2); the remaining 11 children did Set A first, and then each child sorted the letters in the other set.

Results and Discussion

According to analysis of variance (ANOVA), the 20 cards with dotted letters were sorted faster, $F(1, 22) = 6.05, p < .025$, and more accurately, $F(1, 22) = 7.71, p < .025$, than were the normal letters. The mean sorting time for the dotted letters was 36.7 sec, with a mean error rate of .4%; mean sorting time for the normal letters was 45.5 sec, with a mean error rate of 13.9%.

The letters with the commonly added dot were considerably easier for kindergarten children to distinctively classify than were the normal letters. The relations among attributes, not the attributes themselves, predicted classification performance in this task.

Experiment 2: Context

In this experiment we addressed one aspect of the problem of determining which relations are important to performance. These relations must surely depend on the particular set of stimuli. The added dot in Experiment 1 would not likely aid performance in classifying the *b* (or *p*) were there no other letters perceptually similar to *b* (or to *p*) in the alphabet. A letter should be easy or difficult to classify depending on what other letters are in the set. If there are letters similar to *b* (as are *p*, *d*, and *q*), then *b* may be confused with them; if there are no similar letters, then *b* would be easy for children to learn. Hence, the identically sounded upper case of that letter, *B*, is relatively easy to learn (Gibson et al., Note 1) because there are no letters perceptually similar to it. The context in which a letter occurs—its similarity to other letters—determines whether or not that letter is difficult to identify or to classify. This has been shown to be the case when geometric forms are identified (Monahan & Lockhead, 1977) and classified (Lockhead & King, 1977) by adults. Experiment 2 was designed to determine whether the effect of context on classification performance is also important when the subjects are children and the stimuli are letters.

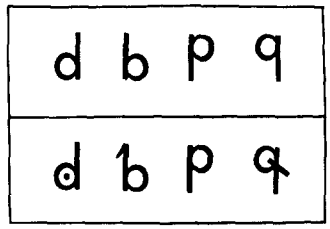

Figure 3. The normal (Set A, top) and the distinctive (Set B, bottom) letters.

Method

Stimuli. The letters studied were the sans serif forms of the lower case letters, *b*, *d*, *p*, and *q*. In the spirit of Experiment 1, a feature was added to each of three of these letters, *b*, *d*, and *q*; the letter *p* was unchanged. In an attempt to magnify the advantage of selected feature relations seen in Experiment 1, a different feature was added here to each different letter: a dot, a left-going slash, and a right-going slash. The normal (N) and distinctive (D) letters used in the study are shown in Figure 3.

Subjects. All members ($n = 27$) of a different kindergarten class from that tested in Experiment 1 were tested individually in early November on a match-to-sample task.

Procedure. The same procedure as described in Experiment 1 was used, with the exception of the cards used in the experimental conditions. Following the warm-up task with the letters *z*, *k*, *l*, and *m*, which each child performed without difficulty, the child did the same match-to-sample task as was used in Experiment 1, but now the child used the letters *p*, *b*, *d*, and *q*.

There were five cards of each of the four letters in the deck, which was thoroughly shuffled before each task. There were two different tasks given to each child. In the N-N task, the cards that were being sorted and the sample cards were all normal sans serif letters; in the D-D task, the cards were all distinctive letters (see Figure 3). Thirteen children performed each task twice, in the order N-N, D-D, N-N, D-D. The remaining children had the sequence D-D, N-N, D-D, N-N.

Results

According to ANOVAs, the tasks were reliably different in sorting times, $F(1, 26) = 7.13, p < .025$, and in errors, $F(1, 26) = 45.07, p < .001$. The mean sorting time for the distinctive letters was 43.7 sec, with a mean of .35 errors per sort. The mean sorting time

for the sans serif letters was 56.2 sec, with a mean of 6.72 errors per sort.

There were 363 errors made by the class when sorting normal letters. Only 19 errors were made when distinctive letters were sorted, and 12 of these 19 errors were made by one child the last time she did the D-D task; she appeared tired then. When this child was retested 3 days later, she continued to make numerous errors on the normal letters, but she made no mistakes using the distinctive letters.

The effects of context on performance can be examined by observing performance on the letter *p*, which had the identical form in the two sets of letters. With the normal deck of cards, there were 94 times when the children erroneously placed cards with the letter *p* into a bin labeled with the letter *b*, *q*, or *d*; there were 89 times when they placed cards with the letter *b*, *d*, or *q* into the bin labeled with the *p*. With the distinctive deck, the same letter *p* was included with the distinctive set of letters, but the children only made 4 errors of placing cards with the letter *p* into the wrong bin, and only 4 times did they place a card with the letter *b*, *d*, or *q* into the bin labeled with the letter *p*. Thus, there were 183 errors associated with the letter *p* when it was a member of the normal letter set and 8 such errors when *p* was a member of the distinctive letter set, producing a factor of 23 : 1 in favor of the distinctive set.

Discussion

The context in which the identical letter is embedded greatly affects performance with that letter. Performance is good or poor depending not on the particular object but on the other objects in the set. Indeed, given only 19 total errors versus 363 total errors, the distinctive letters are *considerably* easier for kindergarten children to classify than are normal sans serif letters. The practical conclusion is clear. The production of clean letters, such as the sans serif ones studied here, which are commonly used in the primary grades, does not provide the best set of letters possible for allowing children to classify letters readily.

The context effect was also strong in this experiment. The normal sans serif letter *p* was difficult for the children to classify when it was embedded in the set of normal letters and was easy to classify when it was embedded in the set of distinctive letters. For the purposes of letter classification, and certainly for the purposes of letter naming and reading as well, the particular letter configuration is not important. Rather, the similarity relations among the different letter configurations are important to performance.

Experiment 3: Numerals

A very restricted set of four letters was employed in Experiments 1 and 2. To determine whether the ideas proposed there would generalize to other objects, we essentially replicated Experiment 2 but used different items. The stimuli were four numerals (2, 5, 6, and 9), which are often noted by teachers as being troublesome for children. Figure 4 depicts two sets of these numerals. The numerals 2–5 and 6–9 are less similar to one another in Set B than in Set A. By the arguments in this article, the numerals of Set B should be easier for the children to use then the numerals of Set A.

Method

Subjects. A class of 27 kindergarten children, different from the classes tested in Experiments 1 and 2, was tested in early October and again in late February.

Procedure. The match-to-sample procedure was basically the same as in Experiments 1 and 2, and the same warm-up task was used in this experiment as well. Each child sorted cards with the numbers of Set A and the numbers of Set B once, and they sorted decks of cards with normal letters twice. (The letter tasks served as a control for a study that is not discussed in the present article.) Fourteen children sorted the numerals of Set B first, and 13 children sorted the nu-

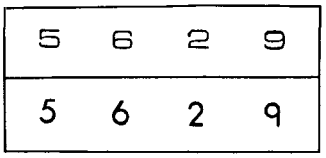

Figure 4. The numerals used in Experiment 3; the letters of Set A (top) are less distinctive than those of Set B (bottom).

Table 1
The Mean Time (sec) and Total Number of Errors Made When Kindergarten Children Sorted Distinctive (D) and Normal (N) Numerals

	Sorting time		Errors	
Testing time	N	D	N	D
October	43.8	38.1	6	1
February	40.7	36.0	1	0

merals of Set A first. One letter task was interposed between the two numeral tasks in all cases.

Results and Discussion

The ANOVA of sorting times revealed significant main effects of time of year (October or February), $F(1, 26) = 15.93, p < .001$, and of which set of numbers was being sorted (A or B), $F(1, 26) = 8.79, p < .01$. The interaction between time of year and numeral set was not significant, $F(1, 26) = .66, p > .25$. The advantage of the distinctive numerals is summarized in Table 1. No formal analysis of the error data was conducted, since the children performed the tasks almost flawlessly, making a combined total of only 8 errors.

This experiment replicated the results of Experiment 2. The type of font used to construct letters and numerals importantly affects the ability of people to classify those items. If letters and numerals in a set are not perceptually similar to one another, then members of that set are easy to classify.

Experiment 4: Matching Letters of Different Fonts: First-Grade Children

If distinctive letters were used in place of the currently popular sans serif letters, the problem of transfer would then become an important issue. It is possible that children can learn to classify distinctive letters quite readily but that this is of no aid in their use of the normal alphabet. Consider, for example, the following suggestion by Crosby (1968) for an Initial Perceptual Alphabet (IPA) designed to avoid visual confusions:

> We are suggesting that a similar technique be used to circumvent some of the visual confusions in the alphabet such as b and d, p and q, m and w, u and v. If another form were substituted for one of the confusable letters—there is no reason that squares, triangles, and other geometric forms cannot be used—it would simplify the visual aspect of reading as the ITA alphabet does the auditory aspect. The child, after benefiting from his IPA exposure, would then be transferred to the regular letter forms. (p. 76)

Although a square and a triangle are certainly less similar to one another than, for example, *b* is to *d*, and are thus less confusable, a potential problem lies in the radically different nature of the symbols that Crosby proposed to substitute for the confusable letters. Squares and triangles are so different from normal letters that it is doubtful whether children would benefit in learning the normal alphabet by first having learned these alternative geometric forms. However, Crosby's intention to reduce confusions may be correct if we can be assured that performance on the new and easily identified alphabet transfers readily to normal type fonts.

The necessary test to determine transfer effectiveness among alphabets is a longitudinal study in which different groups of children are initially taught the distinctive or normal alphabets and later tested on the normal alphabet for reading ability. This study would take several years, and we have not yet attempted to assess the issue of transfer in this manner. However, we offer several observations, some anecdotal and some empirical, which suggest that transfer between distinctive letters constructed in the manner of those of Experiment 2 (see Figure 3, Set B) and letters in more common type fonts, such as those in Figure 1, is not difficult.

Consider the font in which this journal is printed and the font in your daily newspaper. These are very different from the clean type (cf. Figure 1) currently popular in books for teaching young children the alphabet and how to read. Indeed, the letter *d* in Egyptian Number 2 print, with its two serifs, is probably less like the sans serif font *d* than is the dotted *d* in Figure 3. Because children are routinely expected to transfer from clean type to these more ornamental fonts, transferring from our distinctive letters to the fonts adults commonly read would perhaps be no more difficult.

Further anecdotal support comes from our asking 20 first- and second-grade children to name a series of individually presented letters. We instructed the children only that "some of the letters might have some extra marks on them and not to let this bother you." Each distinctive and normal *b, d,* and *q* (Figure 3) was presented once in a randomly arranged series of letters. Half of the children misidentified at least one normal *b, d,* and *q*. However, there was only one mistake made by the 20 children in identifying these letters in the distinctive font. For example, several children did not correctly identify the sans serif *q* but did identify the distinctive *q*. We can at least conclude that these distinctive letters are not more difficult for young children to identify than are normal letters. Recall that the distinctive letters had never been seen by these children.

The empirical evidence supporting the view that transfer between distinctive and normal fonts would not be very difficult for children comes from data on a match-to-sample task that was done with first-grade children (Experiment 4) and later replicated with second-grade children (Experiment 5). The procedure for these experiments was basically the same as for the previous ones, with the exception that the children were asked to do two additional tasks. In these tasks, the deck of cards, or the sample cards over the bins, were normal letters, whereas the other cards, the deck or sample cards, were distinctive letters. These "mixed" tasks were conducted to help assess how difficult it was for a child to make the association between normal letters and distinctive letters. For this purpose, it was also necessary to slightly modify the apparatus. To assure that only the bin labels, and not a card placed into a bin, were available for the match, the rectangular cut-out board was placed above a large box. When a card was placed into a bin by a child, the card dropped out of view and could not be used as a comparison in later trials.

Method

Subjects. All members of a first-grade class ($n = 24$) from a public school were tested individually in late November on a match-to-sample task (see Experiment 2 for details).

Procedure. Each child performed four different tasks: (a) N-N (sort normal letters into bins labeled with normal letters), (b) D-D (sort distinctive letters into bins marked with distinctive letters), (c) N-D (sort normal letters into bins labeled with distinctive letters), and (d) D-N (sort distinctive letters into bins labeled with normal letters). Each child performed each of the four tasks in a random order. To balance effects of practice and fatigue, each child repeated the four tasks in the reverse order.

Observe that the tasks D-N and N-D cannot be performed as match-to-sample tasks. Only the unaltered *p* matches the sample in these cases. Hence, if these traditional tasks are actually performed by matching all features or attributes of the stimuli, the mixed classification task cannot be accomplished readily. The children were told only that some letters might look different from some others and that they should perform the task as well, and as quickly, as they could.

Results and Discussion

None of the children appeared to have any particular difficulties with the changed alphabet, although it was novel to them. According to ANOVAs, there was a difference between tasks in sorting times, $F(3, 69) = 13.1$, $p < .001$, and in errors, $F(3, 69) = 7.90$, $p < .001$. According to planned *t* tests, sorting distinctive letters into bins marked with distinctive letters was done significantly faster and more accurately than each of the other tasks (all $ps < .001$). None of the remaining three tasks was reliably different from each other in terms of sorting times (all $ps > .05$). The D-N task had significantly fewer errors than the N-N task ($p < .05$). Comparisons of the D-N and N-D tasks revealed no reliable differences in terms of

Table 2
The Mean Time (sec) and Mean Number of Errors per Sort When First- and Second-Grade Children Performed Four Sorting Tasks in Experiments 4 and 5

	Sorting time		Errors	
Task	Grade 1[a]	Grade 2[b]	Grade 1[a]	Grade 2[b]
D into D	36.5	—	.33	—
N into N	46.3	39.2	2.00	.86
N into D	48.7	36.6	1.75	.74
D into N	46.4	35.7	1.29	.79

Note. D = distinctive letters; N = normal letters.
[a] Experiment 4. [b] Experiment 5.

errors, nor did the comparison of the N-D and N-N tasks. Table 2 summarizes the data.

Performance on the letter *p*, which had the same shape in all four tasks, was as follows:

1. *Task N-N.* Cards with the letter *p* were misplaced 24 times into wrong bins; cards of the letters *b, d,* or *q* were misplaced 26 times into the *p* bin.
2. *Task N-D.* Cards with the letter *p* were misplaced 10 times into wrong bins; cards of the letters *b, d,* or *q* were misplaced 29 times into the *p* bin.
3. *Task D-N.* Cards with the letter *p* were misplaced 11 times into wrong bins; cards of the letters *b, d,* or *q* were misplaced into the *p* bin 12 times.
4. *Task D-D.* Cards with the letter *p* were misplaced 4 times into wrong bins; cards of the letters *b, d,* or *q* were misplaced 2 times into the *p* bin.

The results comparing the N-N task with the D-D task replicated those reported earlier in this article. The distinctive letters are easier than normal letters for kindergarten children and for first-grade children as well. This study also replicated the finding that context is important. What makes the letter *p* easy or difficult depends on the set with which it occurs. It is not the particular features of a letter that make it easy or difficult to classify; rather, it is how similar that letter is to others that might occur.

A second result of this study warrants attention. This is the observation that the mixed tasks, N-D and D-N, were not more difficult than the N-N task. Indeed, according to the error analysis, the D-N task was easier than the N-N task. This cannot be if indeed the children performed by matching to sample; the mixed tasks are impossible by that assumption. Recall that each card disappeared from view once the child placed it into a bin. Thus, whenever a child attempted to place the dotted *d* into a matching bin during the D-N task, there simply was no bin labeled with a dotted letter. Likewise, in the N-D task, there was nowhere in which to place the normal *d*, since all of the bins were marked with different things. Yet, these mixed tasks were performed at least as well as the N-N task. This strongly suggests that there may be no difficulties encountered if children are asked to sometimes use one font and sometimes use the other font or to transfer between fonts.

Experiment 5: Matching Letters of Different Font: Second-Grade Children

The observation that the mixed sorting tasks are no more difficult than the normal task was sufficiently important to replicate to insure that the result is real. The fact that placing known and identical things together (N-N task) is not easier than placing novel and different things together (D-N and N-D tasks) is not consistent with several viewpoints in psychology. Since we thought the finding might involve the fact that the first-grade children were not fully familiar with normal letters, we decided to evaluate the mixed sorting task with second-grade children.

Method

Subjects and procedure. Twenty-one second-grade children performed the following classification tasks according to the procedures described for kindergarten and first-grade children: (a) N-N (sort normal letters into bins marked with normal letters), (b) N-D (sort normal letters into bins marked with distinctive letters), and (c) D-N (sort distinctive letters into bins marked with normal letters). Because the D-D task has been well demonstrated as being easy, that task was not repeated here. The order of the three tasks was randomized across subjects. To control for practice and fatigue effects, each child repeated the three tasks in reverse order. All 21 children were of normal reading ability; children diagnosed by the school system as having any specical learning difficulties were not used. Before testing, each child was shown each letter individually and asked to identify it. Without exception, the children identified all the letters accurately, including the distinctive letters that were novel to them.

Results and Discussion

An ANOVA showed a difference in sorting times among the tasks, $F(2, 40) = 4.50, p < .025$. According to planned *t* tests, the D-N task was performed reliably faster than the N-N task ($t = 2.19, p < .05$) and was not reliably different than the N-D task ($t = .96$, $p > .25$). The N-D task was performed numerically faster than the N-N task, but the difference was only marginally reliable ($t = 1.80, .10 > p > .05$). All error rates were low,

and the error data did not reveal any significant effects. The results are summarized in Table 2.

This task was not performed as a literal match to sample. Rather, the children performed in what appeared to be a "double-identification" procedure. According to such a notion, the child looks at the card on top of the deck to be sorted and implicitly asks, "What is this?" If the answer is, "This is a *q*," then the child searches for the bin labeled *q* and places the card into that bin when it is found. In such a procedure, whatever is available to facilitate identification of the card to be sorted can aid performance, and whatever is available to facilitate identifying the bin that is sought can be used and can be effective. The distinctive letters facilitate performance. This occurred with children who had no prior experience with these distinctive letters. We have already seen that performance is facilitated most when this advantage is available to both identifications, as in the D-D task.

We conclude at this point that these distinctive letters may be treated as normal letters by children, and they appear to be easier to use than normal letters. This advantage of distinctive over normal letters might disappear, however, if the subjects were better practiced with normal letters, or if they were excellent in their understanding and knowledge of those letters. We tested this in Experiment 6 by using college students who were very experienced with normal letters and who had no prior experience with these distinctive letters.

Experiment 6: Classification Performance by Adults

This experiment was conducted to assess the generalizability of the previously reported results to people who had approximately 20 years experience with normal letters and no experience with the distinctive letters described in Figure 3.

Method

Subjects and procedure. Six right-handed, paid college students, four females and two males, were tested individually on a letter-identification task. Five different letters, *b, d, p, q,* and *k,* were presented on a Tektronix 611 oscilloscope screen that was approximately 90 cm from the observer in a dimly illuminated and sound-attenuating room. The letters *b, d,* and *q* were written in both the normal and the distinctive font, whereas the *p* and *k* appeared only in the normal font. This created a total set of eight items. Responses were made on a keyboard, with keys arranged comfortably under the thumb and four fingers of the preferred hand on a specially constructed keyboard. The same response was to be made to a given letter whether that letter was written in normal or distinctive font. Thus, since the stimulus letters occurred equally often, the responses *b, d,* and *q* each occurred twice as often as did *p* and *k*. These last two letters were included in the study for a reason independent of the discussion in this section. What is of interest here is the relative performance on the *b, d,* and *q* when presented in the normal font as compared with the distinctive font. Stimulus presentation and data collection were controlled by a computer in an adjacent room. All letters were .8 cm × 2 cm, .5° × 1.3° in visual angle, and were drawn in continuous vector mode; the letters were not dot matrix but looked much like normal pica-type sans serif. Each letter was drawn in less than 36 msec. The marks on the altered letters were always written last on the oscilloscope; to the observer the letter appeared to come on at once.

Subjects became accustomed to the procedure by running through a practice block of 50 trials. This was followed by five data blocks of 160 trials each. The subjects were instructed to respond as quickly as possible but without making errors. The letters were presented in random order with the constraint that each of the 8 stimuli was presented 20 times per block. There was a 5-minute rest between blocks.

Results and Discussion

The mean identification times and errors per block of trials for the letters *b, d,* and *q,* written in the normal and distinctive font styles, were analyzed in two ANOVAs. Analysis of the identification times revealed reliable differences due to font style, $F(1, 5) = 37.03, p < .005$, and to practice, $F(4, 20) = 5.7, p < .005$. The Font Style × Practice interaction was not significant ($p > .10$). The mean time to identify the distinctive letters was 554 msec. The mean time to identify the normal font style letters was 606 msec. The mean response time to each stimulus in each practice block is shown in Figure 5.

The ANOVA of the errors revealed only one main effect, font style, to be marginally reliable, $F(1, 5) = 5.5, p < .10$. The mean error rate per block to trials was .66 errors for distinctive letters and 2.26 errors for the normal font. The pattern of errors mirrors that of response times; distinctive letters are

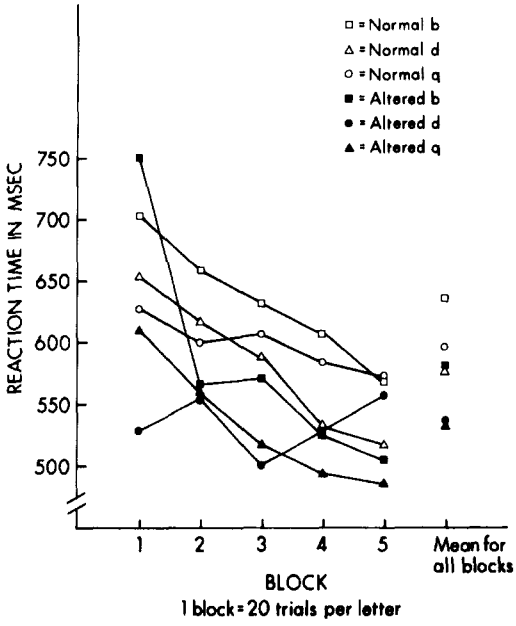

Figure 5. The mean time for college students to classify normal (open figures) and distinctive (filled figures) letters in each of the 5 experimental blocks.

more reliably identified. The lack of significance in the error analysis may be due to a floor effect.

These results with adult subjects are fully consistent with those of children. The distinctive letters really are distinctive, and this is a more important advantage to performance than is 15–20 years of experience with other normal letters. It should be made explicit that the observed differences between letters within fonts (see Figure 5) do not mean that *b*, for example, is difficult compared to *d* and *q*. These differences more likely suggest that responses are easier using some fingers than others; however, the differences in response times between distinctive and normal forms of the same letter are meaningful, because each finger served as its own control in the study.

General Discussion

The distinctive letters are classified more readily than are normal sans serif letters. This is the case for children who are just learning the alphabet (Experiment 2), for those who have recently learned the alphabet (Experiments 4 and 5), and for adults who have been reading for years (Experiment 6).

Although the performance differences between fonts have been reported in this article in terms of sec/sort and error rates, they can also be discussed in terms of years gained. Thus, kindergarten children are faster and more accurate using distinctive letters than are first-grade children using normal letters (43.7 sec and .35 errors per sort vs. 46.3 sec and 2.0 errors per sort). This is a gain of at least 1 year. Likewise, first graders using distinctive letters are faster and more accurate than second graders using normal letters (36.5 sec and .33 errors per sort vs. 39.2 sec and .86 errors) Again, this is a gain of about 1 year. Recall that any second graders with a reading difficulty were selected out of the sample. There were no selections of first-grade or kindergarten children. Thus, this is a conservative estimate of the gain attributed to the distinctive font.

In considering a new font to be used as an aid for children learning to read and to classify, it is necessary that the letters not only be distinctive but that they not interfere with the use of more common fonts. The mixed tasks reported in Experiments 3 and 4 suggest that use of the distinctive letters satisfies this concern. Children can classify nonphysically identical letters together at least as well as they can classify physically identical letters together. The children can also name distinctive letters they had never seen before at least as readily as they can name sans serif letters they had studied for some time.

These conclusions have been based on data collected with normal readers. Distinctive letters provide an even greater advantage for children classified by school systems as *reading disabled* (Crist & Lockhead, Note 2). In that study, reading-disabled children performed the N-N task much more slowly than normal readers, but there was little difference between the two groups when they used distinctive letters (the D-D task). Indeed, when differences in motor skills between the two groups of children were taken into account, it was found that reading-disabled children were faster

and more accurate using distinctive letters than normal readers were using normal letters. Moreover, the D-N task as used in Experiments 4 and 5 in this article proved no more difficult for the reading-disabled children than the N-N task.

Whether the particular distinctive letters employed in the present experiments represent an optimal font to be used with young children is, of course, an open question. Given the approximately 1%–2% error rate found for kindergarten and first-grade children on the speeded D-D task, however, there does not appear to be much room for improvement. The concept involved in the use of distinctive letters is that the readiness with which an item will be distinguished depends on how similar it is to other items in the set. Letters (and other objects, see Lockhead, 1972) are not identified (or confused) in terms of the elements they have or do not have in common. Rather, as Experiment 1 showed, the relations among features determine whether stimuli will be perceived as similar or different from one another. Moreover, as the results of Experiments 2 and 4 showed, it is the relations among objects in the set being judged that determine how easy or difficult any particular stimulus will be to identify and classify. Thus, the identical letter, *p*, was easy or difficult to classify depending on whether it was a member of a set of letters that were perceptually different or similar to it. The particular letter itself is not intrinsically easy or difficult to classify or to identify; relations among letters determine task difficulty.

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